Qualifying Spatial Relations using Jev:

What role could artificial system-one intelligence play in GeoAI?

Disposition for a master thesis by Jakob Roland
Supervised by Prof. Dr. Krzysztof Janowicz
University of Vienna, 2026

Introduction

Since the release of the first chatbot powered by a large language model (LLM) in late 2022, it has become clear that artificial intelligence (AI) has left the realm of science fiction to enter our daily lifes. Since then almost all spheres of society have been impacted in some way by AI, Spatial Data Science being no exception. There have been many works testing LLMs in a variety of spatial applications [EXAMPLES] and some discussing their caveats, challenges, and shortcomings [EXAMPLES]. This thesis sees itself in both categories, recognizing concerns raised over the sustainability of GeoAI [1, 3] and evaluating a new variant of LLM for a spatial task. This thesis tests the spatial understanding of the system-one intelligence model Jev, which was released by the company Typesafe AI in September 2026. To this end, the experiment on spatial relation qualification done by Yuhan Ji (et al, 2025 [2]) is reproduced using Jev instead of ChatGPT-4. In contrast to ChatGPT-4 Jev is quicker, cheaper, and outputs in a fixed format, which facilitates downstream processing of the results. The question remains if the accuracy of Jev's answers is comparable to GPT-4, which this thesis will attempt to find out.